

Highlights

- Historical OD demand ranks road regions for selective exact-index materialization
- Parameter-free bidirectional demand diffusion forms a strong deterministic prior
- A 32-layer residual lifts global Spearman in 12/12 cross-city, cross-window tests
- Budget-aware training cuts shortcuts, keeps online workload within 3% of the prior
- Online answers stay exact; Porto mean query time falls 6.51–7.48% in C++